Mst Afsana Mimi

Amherst, MA, USA | mmimi@umass.edu | +1 207 437 8137 | github.com/mstafsanamimi
linkedin.com/in/mst-afsana-mimi-578234b1/ | mstafsanamimi.github.io

Research Interests

Agentic AI, relational machine learning, database systems, predictive modeling, trustworthy ML, biostatistics, and machine learning for biomedical data analysis.

Current Project

Predictive Query over Relational Databases. Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data. The long-term objective is to build a benchmark for evaluating agentic systems on realistic database tasks.

Education

University of Massachusetts Amherst, Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

- **Advisor:** Lulu Kang
- **Project: Gaussian Process Regression with Fairness Constraint** The project focuses on incorporating fairness constraints into Gaussian Process models that handle both qualitative and quantitative variables to reduce bias in predictions. It aims to ensure fair and reliable performance across protected groups while maintaining predictive accuracy.

University of Dhaka, Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

- **Advisor:** Md Hasinur Rahaman Khan
- **Project: Variable Selection for Censored Data Using Modified Correlation Adjusted Correlation (MCAR) Scores.** Developed an MCAR feature-selection method for censored high-dimensional data using AFT models. Evaluated predictive performance through simulations and DLBCL microarray data.

University of Dhaka, Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Experience

Research Assistant, Data Systems Group, DREAM Lab, Manning College of Information and Computer Sciences, University of Massachusetts Amherst – Amherst, MA, USA. Sep 21, 2026 – Present
Project: *Relational Predictive Query Processing*.
Supervisor: Marco Serafini.

Adjunct Faculty, Department of Psychological and Brain Sciences, University of Massachusetts Amherst – Amherst, MA, USA. Sep 2026 – Present

Teaching Assistant, Department of Mathematics and Statistics, University of Massachusetts Amherst– Massachusetts, USA. Sep 2023 – May 2026

Research Assistant, Department of Computer Science and Engineering, University of Dhaka – Dhaka, Bangladesh. July 2021 – June 2022
Project: *Development of A Robust Bangla Language Model and a Bangla NLP Server (Funded by Centennial Research Grant, University of Dhaka)*.

Research Assistant, Institute of Statistical Research and Training, University of Dhaka – Dhaka, Bangladesh. July 2018 – June 2019
Project: *Poverty and Inequality Mapping in Bangladesh*.

Technical Skills

Programming Languages: Python, R, C, SQL
Frameworks & Libraries: PyTorch, Torch Geometry
Statistical Software: SPSS, Stata, SAS
Data Skills: Subgraph Mining (gSpan), Association Rule Mining

Awards and Scholarship

- **Conference Award for Scientists (CASc)**, International Society for Clinical Biostatistics (ISCB), 2019.
- **National Science and Technology (NST) Fellowship Award** by Education Board of Bangladesh for MS thesis.
- **Dhaka University Merit Scholarship** for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra